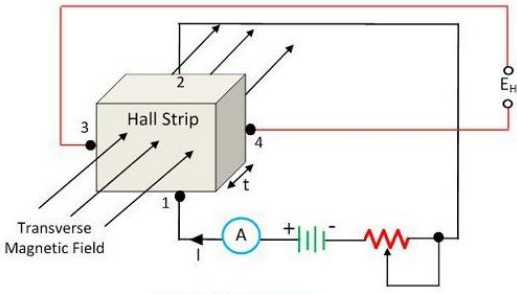
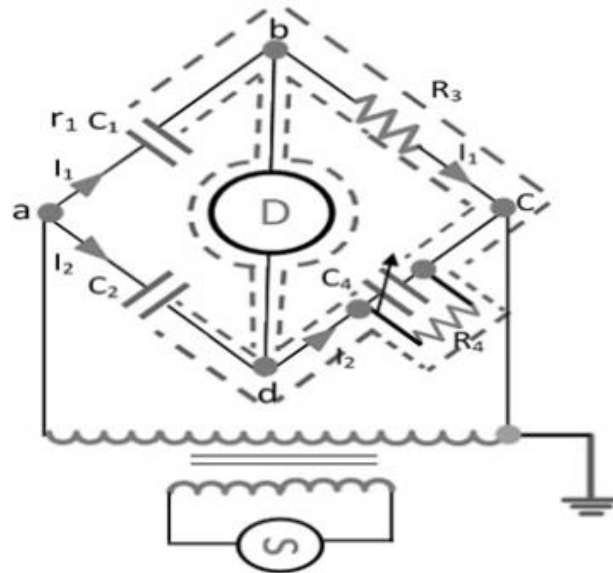


Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY THIRD SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2021		
Course Code: EET203		
Course Name: MEASUREMENTS AND INSTRUMENTATION		
PART A		
		<i>Answer all questions. Each question carries 3 marks</i>
Q1		Define the following terms in measurement i) Accuracy ii) Resolution iii) Precision
Ans		Accuracy The ability of an instrument to measure the accurate value is known as accuracy. In other words, it is the the closeness of the measured value to a standard or true value. Accuracy is obtained by taking small readings. The small reading reduces the error of the calculation Resolution The smallest increment an instrument can detect and display—hundredths, thousandths, millionths. Precision The closeness of two or more measurements to each other is known as the precision of a substance. If you weigh a given substance five times and get 3.2 kg each time, then your measurement is very precise but not necessarily accurate. Precision is independent of accuracy.
Q2		What are the different standards of measurements?
Ans		Different Types of Standards of Measurement are classified in categories 1. International Standards, 2. Primary Standards, 3. Secondary Standards, and 4. Working Standards. International standards are defined by International agreement. They are periodically evaluated and checked by absolute measurements in terms of fundamental units of Physics. The principle function of primary standards is the calibration and verification of secondary standards. Primary standards are maintained at the National Standards Laboratories in different countries. Secondary standards are basic reference standards used by measurement and calibration laboratories in industries. These secondary standards are maintained by the particular industry to which they belong. Secondary standards are basic reference standards used by measurement and calibration laboratories in industries. These secondary standards are maintained by the particular industry to which they belong.
3		Explain the working of Hall effect multipliers.
Ans		The principle of hall effect transducer is that if the current carrying strip of the conductor is placed in a transverse magnetic field, then the EMF develops on the edge of the conductor. The magnitude of

	the develop voltage depends on the density of flux, and this property of a conductor is called the Hall effect. The Hall effect element is mainly used for magnetic measurement and for sensing the current. The metal and the semiconductor has the property of hall effect which depends on the densities and the mobility of the electrons. Consider the hall effect element shown in the figure below. The current supply through the lead 1 and 2 and the output is obtained from the strip 3 and 4. The lead 3 and 4 are at same potential when no field is applied across the strip. 
4	What is a TOD meter?
Ans	Time of the Day (TOD) metering is a technology that enables the implementation of time varying prices under which the price per unit of electricity varies according to the TOD tariff with higher rates at peak periods and low rates at off-peak periods. These time varying tariffs send price signals to consumers that reflect the underlying cost of generating, transmitting and supplying electricity, and enabling resources to be allocated more judiciously and efficiently. Furthermore, price based demand response can reduce or shape consumer demand particularly to reduce load at peak hours on the electricity system. Hence, TOD metering assumes importance in the context of propagating and implementing Demand Side Management (DSM) and achieving energy efficiency.
5	Derive the equation for capacitance using Schering bridge.

Ans



In the figure, 'C1' is the capacitor whose capacitance has to be developed, 'R1' is a series resistance representing the loss in the capacitor C1, C2 is a standard capacitor, 'R3' is a non-inductive resistance, 'C4' is a variable capacitor, and 'R4' is a variable non-inductive resistance in parallel with the variable capacitor 'C4'.

By using the balance condition of the bridge, the ratio of impedance 'Z1 & Z2' are equal to the impedance 'Z3 & Z4', it is expressed as

$$\frac{Z_1}{Z_2} = \frac{Z_3}{Z_4} \quad \text{eq(1)}$$

Where $Z_1 = R_1 + 1/j\omega C_1$; $Z_2 = 1/j\omega C_2$; $Z_3 = R_3$; $Z_4 = (R_4 + 1/j\omega C_4 R_4) / (R_4 - 1/j\omega C_4 R_4)$

Now substitute the values of impedances Z1, Z2, Z3, and Z4 in equation 1, will get the values of C1 and R1.

$$(R_1 + 1/j\omega C_1) [(R_4 + 1/j\omega C_4 R_4) / (R_4 - 1/j\omega C_4 R_4)] = R_3 (1/j\omega C_2) \quad \text{eq(2)}$$

By simplifying the impedance Z4 will get

$$Z_4 = (R_4 + 1/j\omega C_4 R_4) / (R_4 - 1/j\omega C_4 R_4) \\ Z_4 = R_4 / j\omega C_4 R_4 \quad \text{eq(3)}$$

Substitute eq (3) in eq (2) will get

$$(R_1 + 1/j\omega C_1) (R_4 / j\omega C_4 R_4) = R_3 (1/j\omega C_2) \\ (R_1 R_4) + (R_4 / j\omega C_1) = (R_3 / j\omega C_2) (1 + j\omega C_4 R_4)$$

By simplifying the above equation will get

$$(R_1 R_4) + (R_4 / j\omega C_1) = (R_3 / j\omega C_2) + (R_3 * R_4 C_4 / C_2) \quad \text{eq(4)}$$

	Compare real parts R_1 R_4 and $R_3 \cdot R_4 C_4 / 2$ in eq (4) will get unknown resistance R_1 value $R_1 R_4 = R_3 \cdot R_4 C_4 / C_2$ $R_1 = R_3 \cdot C_4 / C_2 \dots \dots \dots \text{eq(5)}$ Similarly compare imaginary parts $R_4 / j\omega C_1$ and $R_3 / j\omega C_2$ will get unknown capacitance C_1 value $R_4 / j\omega C_1 = R_3 / j\omega C_2$ $R_4 / C_1 = R_3 / C_2$ $C_1 = (R_4 / R_3) C_2 \dots \dots \dots \text{eq(6)}$ equations (5) and (6) are the unknown resistance and unknown capacitance
6	How high voltage is tested using sphere gaps?
Ans	A uniform field spark gap will always have a sparkover voltage within a known tolerance under constant atmospheric conditions. Hence a spark gap can be used for measurement of the peak value of the voltage, if the gap distance is known. A sparkover voltage of 30 kV (peak) at 1 cm spacing in air at 20°C and 760 ton pressure occurs for a Sphere Gap Measurement or any uniform field gap. But experience has shown that these measurements are reliable only for certain gap configurations. Normally, only sphere gaps are used for voltage measurements. In certain cases uniform field gaps and rod gaps are also used, but their accuracy is less. The spark gap breakdown, especially the sphere gap breakdown, is independent of the voltage waveform and hence is highly suitable for all types of waveforms from d.c. to impulse voltages of short rise times (rise time $\geq 0.5 \mu\text{s}$). As such, Sphere Gaps are used to measure radio frequency a.c. voltage peak measurements also (up to 1 MHz).
7	Write short notes on RTD.
Ans	An RTD (Resistance Temperature Detector) is a sensor whose resistance changes as its temperature changes. The resistance increases as the temperature of the sensor increases. The resistance vs temperature relationship is well known and is repeatable over time. An RTD is a passive device. It does not produce an output on its own. External electronic devices are used to measure the resistance of the sensor by passing a small electrical current through the sensor to generate a voltage. Typically 1 mA or less measuring current, 5 mA maximum without the risk of self-heating.
8	Describe anyone method of permeability measurement.
Ans	The permeability at several relative orientations between the flow and the preform can be measured using unidirectional and radial methods. In unidirectional test, the liquid is injected from the border of the preform, while in radial tests, the liquid is injected from the center of the preform.
9	Explain the working of electromagnetic flow meter.

Ans	An electromagnetic Flowmeter, simply known as a magnetic flowmeter (Magmeter, Flowmeter, flow measuring device, Megflow, etc.) is a volumetric flow meter. It is a transducer that measures fluid flow by the voltage induced across the liquid by its flow through a magnetic field.Magnetic flowmeters work based on Faraday’s Law of Electromagnetic Induction. The diagram illustrates the principle of an electromagnetic flowmeter. On the left, a cross-section of a pipe is shown with an 'Excitation coil' wrapped around it, an 'Electrode' inside, and 'Measured fluid' flowing through. Labels include 'Lining', 'Inner diameter: D', and 'Pipe'. On the right, a vector diagram shows 'Flow velocity: V' as a blue arrow, 'Magnetic field (magnetic flux density): B' as a green arrow pointing down, and 'Electromotive force: E' as a red arrow pointing right, perpendicular to the other two. According to this principle, when a conductive medium passes through a magnetic field B, a voltage E is generated which is proportional to the velocity v of the medium, the density of the magnetic field, and the length of the conductor. In a magnetic flow meter, a current is applied to wire coils mounted within or outside the meter body to generate a magnetic field. The liquid flowing through the pipe acts as the conductor and this induces a voltage that is proportional to the average flow velocity. This voltage is detected by sensing electrodes mounted in the Electromagnetic flowmeter body and sent to a transmitter which calculates the volumetric flow rate based on the pipe dimensions. Mathematically, we can state Faraday’s law as E is proportional to V x B x L
10	Explain any three types of oscilloscope probes.
Ans	As a fault finding tool for electronics repair, oscilloscope permits users to view the waveforms on the different parts of a circuit. To achieve this, an oscilloscope probe is needed to connect the oscilloscope to the required points. There are two basic types of oscilloscope probes — the active probes and the passive probes. Aside from being cheaper, passive probes are more preferred by oscilloscope users since they can also be used to take a vast variety of measurements. Meanwhile, oscilloscope probes can also be classified in terms of the offered level of attenuation of the signal. Thus, oscilloscope probes are also classified into 100X, 10X, and 1X. The 1X oscilloscope probe is the most basic type. Unlike most probes, it does not attenuate incoming voltage. It also features a length of coax connected to the probe itself and a connector, which is usually a BNC connector, to interface to the oscilloscope. A 10X oscilloscope refers to a probe with an integrated attenuator that delivers an attenuation of 10. This

	allows the circuits' impedance to be enhanced by a factor of 10. While the 10X probe is attenuating the signal, it can also reduce the signal entering the oscilloscope.
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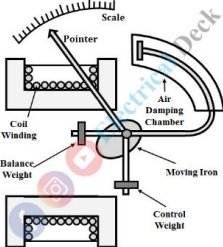
PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

Q11a	Explain with neat diagram the construction and principle of operation of a PMMC instrument. Derive the expression for deflection.
Ans	The term PMMC is the short form of “permanent magnet moving coil”. This instrument is simple as well as most frequently used on ships with sophisticated names. These instruments are used when an exact measurement is required as well as to aid while maintaining electrical equipment. Apart from PMMC, it is also called as D’alvanometer. It is a kind of galvanometer that works on the principle of D’Arsonval. Working Principle of a PMMC Instrument Whenever a current carrying conductor is located within a magnetic field, then it experiences a force that is perpendicular to the current & the field. Based on the rule of “Fleming left hand”, if the thumb, middle and forefinger are at 90 degrees with each other. After that the magnetic field will be in the forefinger, flow of current will be across the middle finger and finally, the force will be through the thumb finger. Once the current flow within the coil on the aluminum former, the magnetic field can be generated in <u>the coil</u> in proportion to the current flow. The electromagnetic force throughout the fixed magnetic field from the permanent magnet generates the deflection force within the coil. After that the spring generates the force to resist additional deflection; therefore it helps to balance the pointer. So damping force can be generated within the system through the aluminum core movement of the magnetic field. It maintains the pointer stable to a point. Once it attains equilibrium by controlling & deflection torque to provide accuracy in measurement. PMMC Instrument Construction The construction of the PMCC instrument can be done using several parts where the permanent magnet and moving coils are essential parts. Each part of this instrument is discussed below.

	<i>Moving Coil</i> It is an essential component of the PMMC instrument. The designing of this coil can be done by winding copper coils to a rectangular block among the magnetic poles. It is made with Aluminum and the rectangular block can be called Aluminum former rotated into the jeweled bearing. So it permits the coil to turn freely. Once the current is supplied throughout these coils, then it gets a deflection within the field, then it is used to find out the voltage or current magnitude. The aluminum is a non-metallic former, used to measure the current whereas the metallic former including high electromagnetic damping is used to calculate the voltage. <i>Magnet System</i> The PMMC instrument includes two high-intensity magnets otherwise a ‘U’ shaped magnet-based design. The designing of these magnets can be done with Alnico & Alcomax for higher superior field intensity & coercive force. In several designs, an extra soft iron cylinder can be arranged among the magnetic poles to create the field identical; while decreasing air reluctance for increasing the strength of the field. <i>Control</i> In the PMMC device, the torque can be controlled due to the springs which are fabricated with phosphorous bronze. These springs are arranged among the two jewel bearings. The spring provides the lane to the lead current to supply in & out of the moving coil. The torque can be controlled mainly due to the delay of the ribbon. <i>Damping Torque</i> Damping torque can be generated within the PMMC instrument using the aluminum core’s movement within the magnetic field. So the pointer can be kept at rest after the early deflection. It assists in the right measurement devoid of fluctuations. Because of the movement of the coil within the magnetic field, eddy current can be generated within the aluminum former. This generates the damping force otherwise torque to resist the motion of the coil. Gradually the deflection of the pointer will be reduced and lastly, it will stop at a permanent position. <i>Pointer and Scale</i> In this instrument, the connection of the pointer can be done through the moving coil. It notices the moving coil’s deflection. The magnitude of their derivation can be displayed on the scale. The pointer within the instrument can be designed with lightweight material. Thus, it can be simply deflected through the coil’s movement. Sometimes, the parallax error can occur within the device which is simply decreased by properly arranging the point. The equation involved in the PMCC instrument is the torque equation. The deflecting torque induces due to the coil’s movement and this can be expressed with the equation shown below. $T_d = NBLdl$
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	Where, 'N' is the no. of turns in the coil 'B' is the density of flux within the air gap 'L' & 'd' are vertical as well as horizontal lengths of the surface 'I' is the flow of current in the coil $G = NBLd$ The restoring torque can be provided to the moving coil can be done with the spring and it can be expressed as $T_c = K\theta$ ('K' is the spring constant) Final deflection can be done through the equation $T_c = T_d$ Substitute the values of T_c & T_d in the above equation, then we can get $K\theta = NBLdI$ We know that $G = NBLd$ $K\theta = GI$ $\theta = \frac{GI}{K}$ $I = \left(\frac{K}{G}\right) \theta$ From the above equation, we can conclude that the deflection torque can be directly proportional to the flow of current in the coil.
b	Derive the expression for deflection for spring controlled attraction type moving iron instrument. Also explain the type of damping provided in moving iron instruments.
Ans	It consists of a fixed coil that is flat with a narrow opening in it. A moving iron that is made of soft iron is mounted on a spindle. The coils are wound with a number of turns that depend upon the range of the instrument. The pointer is mounted on a spindle which consists of a graduated scale for showing the deflection. The construction of attraction type moving iron is shown below.  The controlling torque is provided by the springs or if the instrument is vertically operated gravity control can also be employed. This instrument uses air friction damping to damp out oscillations which consist of a movable piston made of aluminum placed in an air chamber. Since the operating magnetic field produced by the coil winding is not much strong, the eddy current damping which uses permanent magnets can distort the main field. Thus eddy current damping cannot be used and fluid friction damping is not much preferred. The moving iron is made of sheet metal for obtaining a uniform scale.

Let,

T_d = Deflection torque in N-m

θ = Deflection in radians

L = Inductance in Henry

I = Initial current

dI = Change in initial current

dL = Change in inductance

$d\theta$ = Change in deflection

The voltage drop across the ammeter is given by,

$$e = \frac{d}{dt}(LI) = L \frac{dI}{dt} + I \frac{dL}{dt}$$

The energy consumed by the meter,

$$E = e I dt = L \frac{dI}{dt} I dt + I \frac{dL}{dt} I dt$$

$$= LI dI + I^2 dL \dots (1)$$

For a small increment in current (dI), the deflection is increased by $d\theta$. We know that the energy stored by the inductance of the meter = $\frac{1}{2} LI^2$. Therefore, new energy storage due to increment in current,

$$= \frac{1}{2} (L + dL)(I + dI)^2$$

The change in stored energy is,

$$= \frac{1}{2} (L + dL)(I + dI)^2 - \frac{1}{2} LI^2$$

$$= \frac{1}{2} (L + dL)(I^2 + 2IdI + dI^2) - \frac{1}{2} LI^2$$

$$= \frac{1}{2} LI^2 + LI(dI) + \frac{1}{2} L(dI^2) + \frac{1}{2} I^2(dL)$$

$$+ I(dL)(dI) + \frac{1}{2} (dL)(dI^2) - \frac{1}{2} LI^2$$

Neglecting the second and third order differential terms, we get change in stored energy as,

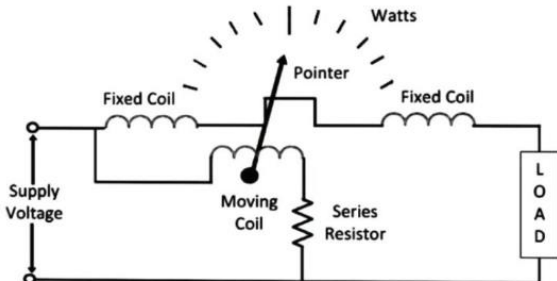
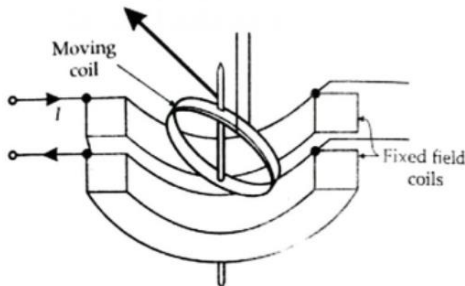
$$= LI dI + \frac{1}{2} I^2 dL \dots (2)$$

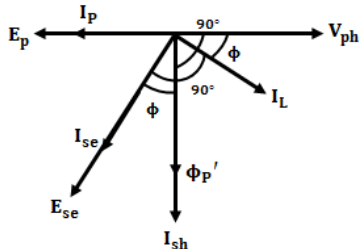
We have work done =

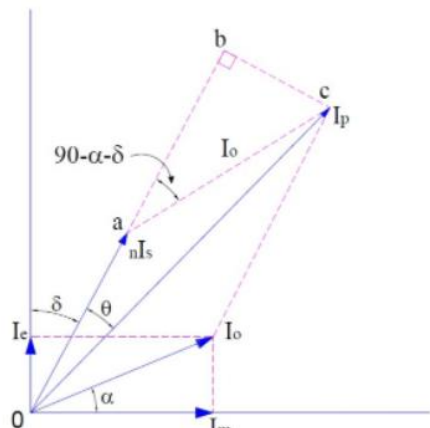
	$T_d \times d\theta = T_d d\theta \dots(3)$ From the law of conservation of energy, we have, Energy consumed = Work done + Change in stored energy Substituting equations 1, 2, and 3 in 4, we get, $LI dI + I^2 dL = T_d d\theta + LI dI + \frac{1}{2} LI^2$ $T_d d\theta = \frac{1}{2} I^2 dL$ $\therefore T_d = \frac{1}{2} I^2 \frac{dL}{d\theta}$ From the above, we can say deflecting torque produced is proportional to the square of the current. For ac measurement, the readings obtained will be the RMS value of the current or voltage. Due to square law response of the instrument, the scale of moving iron instruments is not uniform.
Q12 a	What is the different torques needed for proper operation of an indicating instrument?
Ans	The torques are: 1. A deflecting or operating torque, T_d 2. A controlling torque, T_c 3. A damping torque, T_v. The deflecting torque, causes the moving system of the instrument to move from its zero position. The controlling torque is dependent on the magnitude of deflection produced. The moving system is deflected from zero to such a position that the controlling torque at that deflected position is equal to the deflecting torque. The damping torque is proportional to the speed of rotation of the moving system
b	The coil of a measuring instrument has a resistance of 1Ω and the instrument has a full scale deflection of $250V$ when a resistance of 4999Ω is connected in series with it. Find I. The current range of the instrument when used as an ammeter with the coil connected across a shunt of $1/499\Omega$. II. The shunt resistance for the instrument to give a full scale deflection of $50A$.
Ans	I) $V=IR$ Current in the series circuit $I = \frac{250}{4999+1} = 0.05A$

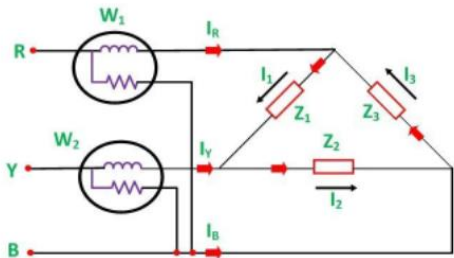
	Current through shunt $= \frac{I_m R_m}{R_{sh}} = \frac{0.05 \times 1}{\frac{1}{499}} = 24.95 \text{ A}$ Current range of instrument $= 24.95 + 0.05 = 25 \text{ A}$ II) Current through shunt $= \frac{I_m R_m}{R_{sh}} = \frac{0.05 \times 1}{R_{sh}} = 50$ $R_{sh} = \frac{0.05}{50} = 1/1000 \Omega$
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Module 2

Q13a	With neat diagram explain the construction and working of an electrodynamicometer wattmeter.
Ans	Working Principle The Electrodynamometer wattmeter working principle is very simple and easy. It is based on the theory of a current-carrying conductor experiences a magnetic force when it is placed in a magnetic field. Hence there will be a deflection of pointer that took place due to the mechanical force. It contains two coils such as fixed coil (current coil) and moving coil (pressure coil or voltage coil). The fixed coil is used to carry the current and connected in series with the load in any circuit. The moving coil carries the current directly proportional to the voltage and connected across the voltage. The value of current limited to a minimum value due to large non-inductive resistance connected in series. The circuit diagram is shown below.  Construction of Electrodynamicometer Wattmeter The construction of the Electrodynamicometer wattmeter includes fixed coil, moving coil, control, damping, scales, and pointer. The construction of the Electrodynamicometer wattmeter is shown below. 

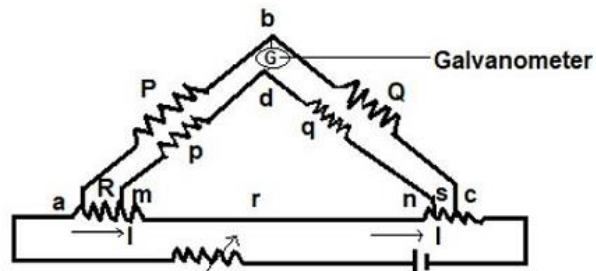
	Fixed Coil It is connected in series with the load, which is considered as the current coil. To make construction easy and simple, it is divided into two parts. Those are two elements connected parallel to each other. It produces a uniform electric field, which is very essential for working. The current coil is designed in such a way that it carries approximately 20 Amperes. Moving Coil Considered as pressure coil in this instrument, that is connected parallel with the supply voltage. So, that current flows directly proportional to supply voltage. A pointer is mounted on the moving coil with the help of spring to control the movement. The temperature increases when current flows through the coil. So, in order to control the flow of the current resistor is connected in series with the moving coil. Control It provides controlling torque onto the instruments. Gravity control and spring control are the two types in this control system. Among these two Electrodynamometer wattmeters uses a spring control system as it helps in the pointer movement. Damping The effect which reduces the pointer movement is called damping. In this, damping torque is produced because of the air friction. Other types of damping are not used as they destroy the useful magnetic flux. Scales and Pointers It uses a linear scale as the moving coil moves linearly. The apparatus uses knife-edge pointers in order to remove parallax error caused due to oversights
13b	Derive the expression for torque of a single phase induction type energy meter.
Ans	The phasor diagram of the energy meter is shown below.  <i>Therefore, average torque is given as,</i>

	$T \propto \phi_p I_{se} \cos \phi - \phi_{se} I_p \cos(180^\circ - \phi)$ $T \propto \phi_p I_{se} \cos \phi + \phi_{se} I_p \cos \phi$ $T \propto \cos \phi (\phi_p I_{se} + \phi_{se} I_p)$ But, $I_p \propto \phi_p \propto V_{ph}$ $I_{se} \propto \phi_{se} \propto I_L$ From this, $\phi_p I_{se} \propto V_{ph} I_L \Rightarrow \phi_p I_{se} = A V_{ph} I_L$ <i>and</i> $\phi_{se} I_p \propto V_{ph} I_L \Rightarrow \phi_{se} I_p = B V_{ph} I_L$ $\therefore T \propto (A V_{ph} I_L + B V_{ph} I_L) \cos \phi$ $T \propto (A + B) V_{ph} I_L \cos \phi$ Since $A + B$ is a constant, $T \propto V_{ph} I_L \cos \phi$ From the above, the average torque produced in the disc is proportional to the actual power consumed in the load.
14a	Derive the expression for ratio and phase angle error in a current transformer.
Ans	Phase Angle of Current Transformer: Phase angle of current transformer is the angle between the primary current I_p and secondary current I_s when reversed. In above phasor diagram, θ is the phase angle.  In the above phasor, consider right angled triangle obc $\tan \theta = bc/ob$ $= I_0 \sin(90 - \alpha - \delta) / (oa + ab)$

	$= I_0 \cos(\alpha + \delta) / [nI_s + I_0 \sin(\alpha + \delta)]$ Since θ is very small, hence $\tan \theta$ will be approximately equal to θ. Therefore we can write, $\Theta = I_0 \cos(\alpha + \delta) / [nI_s + I_0 \sin(\alpha + \delta)]$ Also, I_0 is very very small, therefore $I_0 \sin(\alpha + \delta) \ll nI_s$, neglecting $I_0 \sin(\alpha + \delta)$ $\Theta = I_0 \cos(\alpha + \delta) / nI_s$ $= [I_0 \cos \alpha \cos \delta - I_0 \sin \alpha \sin \delta] / nI_s$ From phasor, $I_m = I_0 \cos \alpha$ and $I_e = I_0 \sin \alpha$, hence $\Theta = [I_m \cos \delta - I_e \sin \delta] / nI_s \text{ Radian}$ $= (180/\pi) [I_m \cos \delta - I_e \sin \delta] / nI_s \text{ degree}$ Ratio Error of Current Transformer: Ratio Error is defined as the per unit deviation in transformation ratio from nominal ratio. It is expressed in percentage. Percentage Ratio Error E $= (\text{Nominal Ratio} - \text{Transformation Ratio}) \times 100 / \text{Transformation Ratio}$ Since the burden of current transformer is usually resistive, therefore power factor of burden is almost unity. Hence $\delta = 0$, $R = n + (I_0 \sin \alpha / I_s) \text{ but } I_0 \sin \alpha = I_e$ $= n + I_e / I_s$ Therefore, percentage ratio error can be calculated easily.
14b	Explain how power can be measured in a 3-phase circuit using two wattmeters with a neat circuit diagram in unbalanced load condition.
Ans	Considering the delta connected circuit shown in the figure below:  The instantaneous current through the coil of the wattmeter, W_1 is given by the equation: $W_1 = i_R = i_1 - i_3$ Instantaneous power measured by the Wattmeter, W_1 will be: $W_1 = e_{RB}$ Therefore, the instantaneous power measured by the wattmeter, W_1 will be given as:

	$W_1 = e_{RB} (i_1 - i_3) \dots \dots \dots (3)$ The instantaneous current through the current coil of the Wattmeter, W_2 is given as: $W_2 = i_Y = i_2 - i_1$ The instantaneous potential difference across the potential coil of wattmeter, W_2 : $W_2 = e_{YB}$ Therefore, the instantaneous power measured by Wattmeter, W_2 will be: $W_2 = e_{YB} (i_2 - i_1) \dots \dots \dots (4)$ Hence, to obtain the total power measured by the two wattmeter the two equations, i.e. equation (3) and (4) has to be added. $W_1 + W_2 = e_{RB} (i_1 - i_3) + e_{YB} (i_2 - i_1)$ $W_1 + W_2 = i_1 e_{RB} + i_1 e_{YB} - i_3 e_{RB} - i_1 e_{YB}$ $W_1 + W_2 = i_2 e_{YB} + i_3 e_{BR} - i_1 (e_{YB} + e_{BR}) \quad (\text{i.e. } -e_{RB} = e_{BR})$ $W_1 + W_2 = i_1 e_{RY} + i_2 e_{YB} + i_3 e_{BR} \quad (\text{i.e. } e_{RY} + e_{YB} + e_{BR} = 0)$ $W_1 + W_2 = P$ Where P is the total power absorbed in the three loads at any instant. The power measured by the Two Wattmeter at any instant is the instantaneous power absorbed by the three loads connected in three phases. In fact, this power is the average power drawn by the load since the Wattmeter reads the average power because of the inertia of their moving system.
15 a	With the help of diagram explain the working of Kelvin double bridge.
Ans	Definition: A kelvin bridge or kelvin double bridge is a modified version of the Wheatstone bridge, which can measure resistance values in the range between 1 to 0.00001 ohms with high accuracy. It is named because it uses another set of ratio arms and a galvanometer to measure the unknown

resistance value. The basic operation of the Kelvin double bridge can be understood from the basic construction and operation of the kelvin bridge. The construction of Kelvin double bridge is similar to the wheat stone bridge, but the only difference is it consists of 2 arms “P & Q”, “p & q” where the arm “p & q” is connected to one end of the galvanometer, at “d” and “P & Q” is connected to another end of the galvanometer, at ‘b’. This connection minimizes the effect of connecting lead and the unknown resistor R & a standard resistor S is placed between ” m and n”, and “a and c”.



Derivation

The ratio $p/q = P/Q$,

Under the balanced condition current in galvanometer = 0

Potential difference at a & b = voltage drop between Eamd .

$$E_{ab} = [P / P+Q] E_{ac}$$

$$E_{ac} = I [R + S + [(p+q)r] / [p+q+r]] \dots\dots\dots(3)$$

$$E_{amd} = I [R + (p / (p+q)) * \{(p+q) r / (p+q+r)\}]$$

$$E_{ac} = I [p r / (p+q+r)] \dots\dots\dots(4)$$

When galvanometer shows zero then

$$(P / P+Q) * I [R + (p / (p+q)) * \{(p+q) r / (p+q+r)\}] = I [p r / (p+q+r)]$$

$$R = (P / R) * S + p r / (p+q+r) [(P/Q) - (p/q)]$$

We know that $P/Q = p/q$

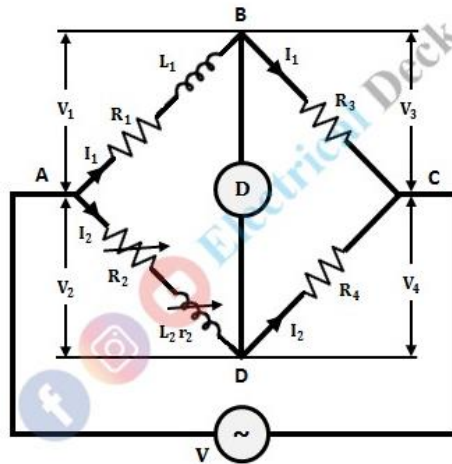
$$R = (P/Q) * S \dots\dots\dots(5)$$

For obtaining perfect results, the arms ratio should be maintained equal and the thermo-electric electromagnetic field induced in the bridge while taking readings can be reduced by interchanging the polarity of the connection. Therefore the unknown resistance value can be obtained from the two arms. Typically, it measures $1 - 0.00001$ ohm with an accuracy $\pm 0.05\%$ to $\pm 0.2\%$, in order to achieve sensitivity the current to be supplied should be large.

15b Draw the circuit and phasor diagram of Maxwell's Inductance bridge and derive the expression for unknown inductance.

Ans **Maxwell's Inductance Bridge :**

In Maxwell's inductance bridge the unknown self-inductance to be measured is compared with the known inductance. Hence the unknown self-inductance and internal resistor of an inductor can be measured with Maxwell's inductance bridge. The circuit diagram is shown in the below figure.



Let,

L_1 = Unknown inductance to be measured

R_1 = Resistance of the Unknown inductance

R_3 , & R_4 = Standard non-inductive resistances

R_2 = Standard Variable resistance

L_2 = Standard Variable inductance with fixed resistance r_2 .

From the above figure, the impedances of the respective arms are given as,

$$Z_1 = (R_1 + j\omega L_1)$$

$$Z_2 = (R_2 + r_2 + j\omega L_2)$$

$$Z_3 = R_3$$

$$Z_4 = R_4$$

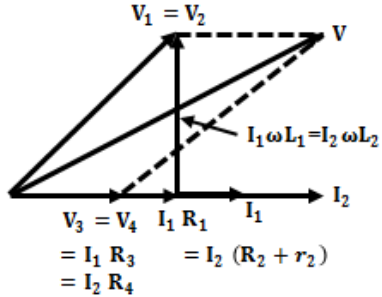
Under balanced condition (i.e., when detector shows null deflection), we have,

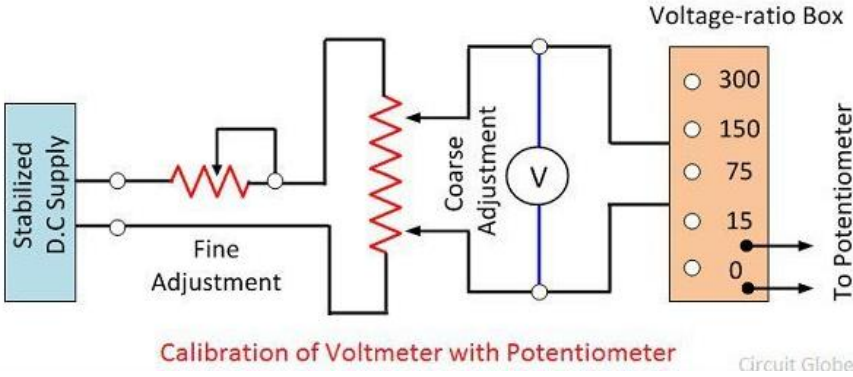
$$Z_1 Z_4 = Z_2 Z_3$$

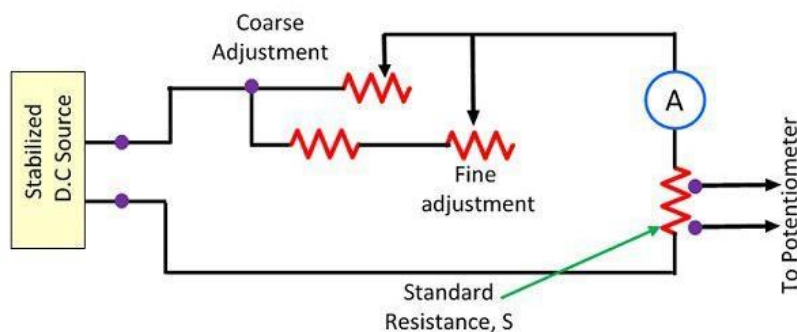
$$(R_1 + j\omega L_1) \times R_4 = (R_2 + r_2 + j\omega L_2) \times R_3$$

$$R_1 R_4 - R_2 R_3 - r_2 R_3 + j\omega(L_1 R_4 - L_2 R_3) = 0$$

On equating the real and imaginary parts on both sides, we get,

	$R_1 R_4 - R_2 R_3 - r_2 R_3 = 0$ $R_1 R_4 = R_2 R_3 + r_2 R_3$ $R_1 = \frac{R_3}{R_4} (R_2 + r_2)$ Also, $L_1 R_4 - L_2 R_3 = 0$ $L_1 R_4 = L_2 R_3$ $L_1 = L_2 \times \frac{R_3}{R_4}$ <i>Hence, the unknown self-inductance and resistance of the inductor are obtained in terms of known standard values. Also, both the equations are independent of frequency term.</i> Phasor Diagram of Maxwell's Inductance Bridge : In Maxwell's inductance bridge, the resistance R_2 is the decade resistance box. The resistor R_3 and R_4 can have their values as from 10, 100, 1000, and 10,000 Ω by the successive adjustment of L_2 and R_2. The balance of the bridge can be adjusted. The phasor diagram of Maxwell's inductance bridge under balance condition is shown below  Phasor Diagram of Maxwell's Inductance Bridge
16a	Explain the working of a Meggar with the help of a neat diagram.
Ans	Definition: The Megger is the instrument uses for measuring the resistance of the insulation. It works on the principle of comparison, i.e., the resistance of the insulation is compared with the known value of resistance. If the resistance of the insulation is high, the pointer of the moving coil deflects towards the infinity, and if it is low, then the pointer indicates zero resistance. The accuracy of the Megger is high as compared to other instruments. Working of Megger The testing voltage is usually 500, 1000 or 2500 V which is generated by the hand driven generator. The generator has centrifugal clutch due to which the generator supplied the constant for the

	insulation test. The constant voltage is used for testing the insulation having low resistance. The Megger has three coils two pressure coils and one current coil. The pressure coil rotates the moving coil in the anticlockwise direction, whereas the current coil rotates it in the clockwise direction. When the unknown resistance is connected in the circuit, the pointer of the moving coil becomes stable. The pressure coil and the current coil balance the pointer and set it in the middle of the scale. The deflection of the pointer is directly proportional to the voltage applied to the external circuit. When the testing circuit is applied across the Megger, and if there is no shorting throughout the insulation then the pointer deflects towards the infinity. Which shows that the resistance has high insulation. For low resistance, the pointer moves towards zero.
b	What is a DC potentiometer? Explain the calibration of ammeter and voltmeter using it with neat diagrams.
Ans	A DC potentiometer can measure voltage directly and measure resistance, current, power, and temperature indirectly. A DC potentiometer is a potentiometer in which the supply is a battery and the balance is under direct current conditions. Calibration of Voltmeter The circuit for the calibration of the voltmeter is shown in the figure below.  The circuit requires two rheostats, one for controlling the voltage and another for adjustment. The voltage ratio box is used to step-down the voltage to a suitable value. The accurate value of the voltmeter is determined by measuring the value of the voltage to the maximum possible range of the <u>potentiometer</u>. The potentiometer measures the maximum possible value of voltages. The negative and positive error occurs in the readings of the voltmeter if the readings of the potentiometer and the voltmeter are not equal. Calibration of Ammeter The figure below shows the circuit for the calibration of the ammeter.



Calibration of an ammeter with Potentiometer

The standard resistance is connected in series with the ammeter which is to be calibrated. The potentiometer is used for measuring the voltage across the standard resistor. The below mention

$$I = \frac{V_s}{S}$$

formula determines the current through the standard resistance.

Where,

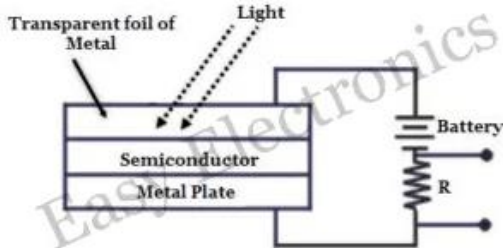
V_s – voltage across the standard resistor as indicated by the potentiometer.

S – resistance of standard resistor

This method of calibration of the ammeter is very accurate because in this method the value of standard resistance and the voltage across the potentiometer is exactly known by the instrument.

17a Explain the method of flux measurement using ballistic galvanometer.

Ans A galvanometer is similar to a sensitive ammeter, differing mainly in that when no current passes through the meter, the needle is in the middle of the dial rather than at the left hand end. A galvanometer is used not so much to *measure* a current, but rather to *detect* whether or not a current is flowing, and in which direction. In the *ballistic* galvanometer, the motion of the needle is undamped, or as close to undamped as can easily be achieved. If a small quantity of electricity is passed through the ballistic galvanometer in a time that is short compared with the period of oscillation of the needle, the needle will jerk from its rest position, and then swing to and fro in lightly damped harmonic motion. (It would be simple harmonic motion if it could be completely undamped.) The amplitude of the motion, or rather the extent of the first swing, depends on the quantity of electricity that was passed through the galvanometer. It could be calibrated, for example, by discharging various capacitors through it, and making a table or graph of amplitude of swing versus quantity of electricity passed. The coil of a moving-coil meter is wound around a small aluminium frame called a *former*. When the current through the ammeter coil is changed, the coil – and the former – swing round; but a current is induced in the former, which gives the former a magnetic moment in such a sense as to *oppose* and therefore dampen the motion. The resistance of the former is made just right so that critical damping is achieved, so that the needle reaches its equilibrium position in the least time without overshoot or swinging. The little aluminium former does not look as if it were an important part of the instrument – but in fact its careful design is very important!

b	Describe the construction and working of photoconductive and photovoltaic cells.
Ans	Photoconductive cell construction and working - The photoconductive cell (PC) is a two-terminal light-sensitive semiconductor device. - The semiconductor materials are made in the form of zig-zag strips with their ends being attached to external pins. The whole assembly is enclosed in a glass cover or protected in transparent plastic. - Cadmium sulfide (CdS) and cadmium selenide (CdSe) are the materials that are normally used in photoconductive cell manufacture. - Nowadays, other materials like lead selenide and lead telluride are also used in the manufacture of photoconductive cells. The simple circuit for a photoconductive cell is shown in the figure below:  - In the circuit shown, the resistance of the photoconductive cell is in series with an external resistance(R) which limits the current. A bias voltage is also used in series with the photoconductive cell. - In the absence of light, the resistance of the photoconductive cell is very high and hence the current as measured using an ammeter is also very low. This current is called dark current and the resistance corresponding to this is called dark resistance. - This reduction in current leads to a voltage drop across the load resistor R. - When the photoconductive cell is exposed to external light, the resistance of the photoconductive cell decreases, resulting in a large current flow called the joint current. This is because the light energy supplied to the photoconductive cell causes its covalent bonds to be broken, resulting in the creation of electron-hole pairs - Due to this increase in the number of charge carriers, the resistance of the material is proportional to the number of charge carriers. Construction of Photovoltaic cell The photovoltaic cell is a semiconductor pn junction device. However, its construction is not the same as a normal junction diode. It is formed by a combination of p-type semiconductor material with an n-type semiconductor. Usually, silicon and selenium are used as the basic semiconductor material. However, gallium arsenide, cadmium sulphide are also majorly used.

The reason why we have constructed the two regions differently is that the region from where **the light ray is allowed to incident must be thin**. So, that the incident radiation can easily penetrate to the depletion region (pn junction).

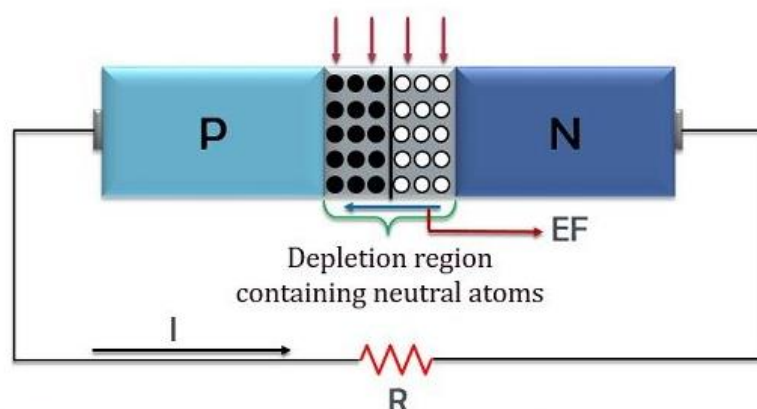
A layer of glass is placed at the top surface of the device in order to gather maximum incident radiation. Also, metallic fingers are provided at the top surface of the structure. Here, we have used silver fingers. As they are good conductors and absorbs the released electrons easily in order to provide proper conduction.

Working of Photovoltaic cell

As we have already discussed at the beginning of the article that photovoltaic cells or solar cells are **light sensitive devices** that produce an electric voltage or current when its surface is illuminated with radiation.

So, now have a look at how potential difference is generated when no external potential is provided but a light ray is incident on it.

The figure below represents a detailed structure showing the working of the solar cells:



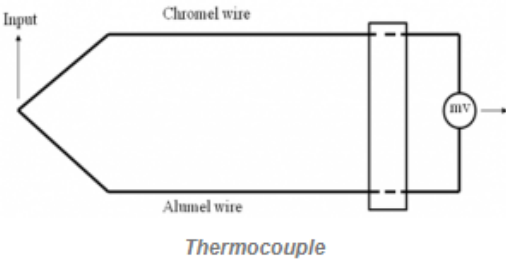
When no any light energy is provided to the device then the device does not conduct. Only the intensity of radiation falling at the surface allows a proportional current to flow through the device.

When a light ray is permitted to incident at the top surface of the structure. Then the glass placed gathers the light energy and permit it to reach the p region. As at the time of construction we have discussed that the thin p region is fabricated in order to have easy light penetration towards the junction region.

18a

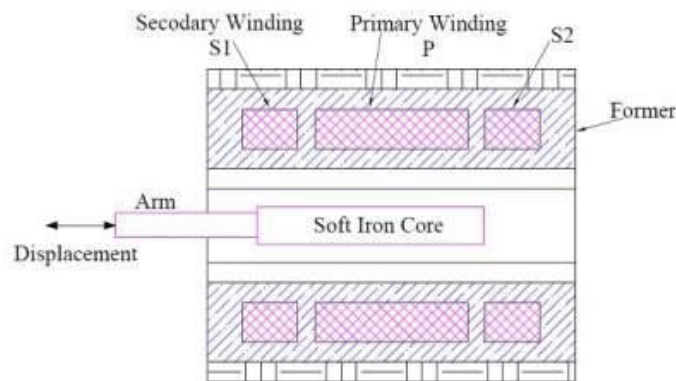
Explain in detail the measurement of iron losses in a magnetic material using Llyod-Fisher square wattmeter method.

Ans	Wattmeter method- Lloyd Fisher square This method is used for measurement of iron loss in strip(sheet) material. The strip material to be tested is assembled as a closed magnetic circuit in the form of a square. Therefore this arrangement is known as a magnetic square. Lloyd Fisher square is the most commonly used magnetic square. The strips are usually 0.25m long and 50 to 60 mm wide. Each stack is made up of two types of strips one in the direction of rolling and the other cut perpendicular to the direction of rolling. The individual strips are insulated from each other. These strips are built up into four stacks and assembled to form a complete magnetic circuit with the help of bent corner pieces and clamps. These corner pieces should be of the same material as the strips or atleast material having similar magnetic properties. The overlapping at the corners should be only a few mm, and a correction may be applied if necessary to account for the fact that the cross section of the magnetic circuit is doubled at the overlapping joints. Ref fig.18.20. A very small no of strips are shown in the fig for sake of clarity. The bundles of strip are placed inside four similar magnetizing coils of heavy wire connected in series to form primary winding. Each of these coils has, underneath it, two single layer coils of thin wire having equal no of turns. These secondary coils are connected in series in groups of four – one on each core to form two separate and similar secondary wdgs. Advantage:- If required corrections are made to the measured value, the Lloyd Fisher square is more reliable than Epsom square. The use of corner pieces makes the Lloyd Fisher square superior for testing of anisotropic materials. Test Setup:- Fig18.21 shows the connection diagram for finding total iron loss by wattmeter method. The test specimen is weighed before assembly and its effective cross section is determined. The primary wdg is connected to a sinusoidal voltage supply. The primary wdg contains the cc of the wattmeter. The pc is supplied from one of the secondary winding, S1. The second secondary wdg S2 supplies an electrostatic voltmeter. The supply frequency is adjusted to the correct value and the magnetizing current is adjusted to give the required value of B_{max} and readings of wattmeter and voltmeter are noted. The electrostatic voltmeter connected across S2 measures the rms value of induced voltage. The value of induced voltage is, $E = 4K_f B''_{max} A_s f N_2$ volts Where $K_f \diamond$ Form factor $B''_{max} \diamond$ max flux density(apparent) wb/m² $A_s \diamond$ Cross sectional area of specimen $f \diamond$ supply frequency Hz; $N_2 \diamond$ no of turn of sec wdg S2 The cross section of the coil A_c is larger than A_s. Thus S2 encloses flux in the air space between specimen and coil in addition to the flux in the specimen. Correction need to be applied for the value of B_{max}. Thus, actual value of flux density in the specimen is $B_m = B''_{max} - \mu_0 H_m (-1)$ where $H_m \diamond$ maximum magnetizing force. It is required to find the iron loss of the specimen. The wattmeter reading includes the iron loss of specimen and the copper loss of secondary winding ckt. The copper loss of secondary ckt can be calculated and subtracted from the wattmeter reading. Let $P_i \diamond$ total iron loss; $W_P \diamond$ wattmeter reading; $V \diamond$ voltage applied to wattmeter pc $E \diamond$ voltage reading of voltmeter; voltage induced in S2: volts $r_p \diamond$ resistance of wattmeter pc $r_c \diamond$ resistance of coil S1; ohms $I_p \diamond$ current in pc ckt; A Voltage actually induced in wdg S1 = E (sec no of turns are equal for S1 and S2) ie $E = I_p(r_p + r_c)$ Total copper loss in secondary = $E^2 / (r_p + r_c)$ Total iron loss in specimen + copper loss in sec wdg ckt = PE/V [because, the wattmeter reading is $P = (cc \text{ current} \times pc \text{ voltage}) = I_c \times V$. The pc voltage is reduced due to the resistance drop of sec coil. It should be actually E to give the correct power loss. Thus, actual loss is $I_c \times E$ or $(P/V) \times E = PE/V$] Therefore, total iron loss of specimen $P_i = (PE/V) - E^2 / (r_p + r_c)$ watts Specific iron loss can be calculated by dividing the total iron loss by weight of the specimen Similarly, values of iron loss can be determined for other frequencies and values of P_h and P_e can be separated as explained above.
18b	Explain the temperature measurement using thermocouples.

Ans	The thermocouple can be defined as a kind of temperature sensor that is used to measure the temperature at one specific point in the form of the EMF or an electric current. This sensor comprises two dissimilar metal wires that are connected together at one junction. The temperature can be measured at this junction, and the change in temperature of the metal wire stimulates the voltages.  The amount of EMF generated in the device is very minute (millivolts), so very sensitive devices must be utilized for calculating the e.m.f produced in the circuit. The common devices used to calculate the e.m.f are voltage balancing potentiometer and the ordinary galvanometer. From these two, a balancing potentiometer is utilized physically or mechanically. Thermocouple Working Principle The thermocouple principle mainly depends on the three effects namely Seebeck, Peltier, and Thompson. Seebeck-effect. This type of effect occurs among two dissimilar metals. When the heat offers to any one of the metal wires, then the flow of electrons supplies from hot metal wire to cold metal wire. Therefore, direct current stimulates the circuit. Peltier-effect This Peltier effect is opposite to the Seebeck effect. This effect states that the difference of the temperature can be formed among any two dissimilar conductors by applying the potential variation among them. Thompson-effect This effect states that as two disparate metals fix together & if they form two joints then the voltage induces the total conductor's length due to the gradient of temperature. This is a physical word that demonstrates the change in rate and direction of temperature at an exact position.
19 a	Explain the basic principle and working of LVDT.

Ans

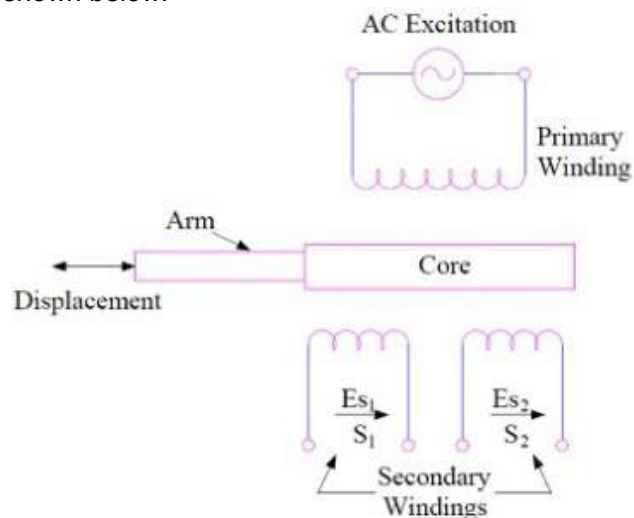
Linear Variable Differential Transformer, LVDT is the most used inductive transducer for converting translating linear motion into electrical signal. This transducer converts a mechanical displacement proportionally into electrical signal. LVDT is a transformer consisting of one primary winding P and two secondary winding S_1 & S_2 mounted on a cylindrical former. The two secondary winding have equal number of turns and placed identically on either side of the primary winding as shown in figure below.



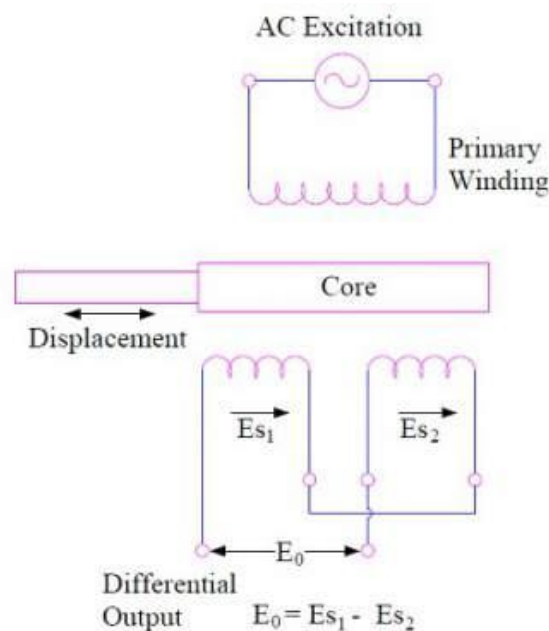
A movable soft iron core is placed inside the former. Actually the movable core is made of nickel iron with hydrogen annealed. Hydrogen annealing is done to eliminate harmonics, residual voltage of core and thus provides high sensitivity. The movable core also is laminated in order to reduce eddy current loss. The assembly of laminated core is placed in a cylindrical steel housing and end lids are provided for electromagnetic and electrostatic shielding. The displacement to be measured is attached to this movable soft iron core.

LVDT- Working Principle:

Since the primary winding of Linear Variable Differential Transformer (LVDT) is supplied with AC supply, it produces an alternating magnetic flux in the core which in turn link with the secondary winding S_1 and S_2 to produce emf due to transformer action. The electrical equivalent circuit of LVDT is shown below.



Let us assume that the emf produced in secondary winding S_1 is E_{s1} and that in S_2 is E_{s2} . The magnitude of E_{s1} and E_{s2} will depend upon the magnitude of rate of change of flux ($d\phi / dt$) as per the Faraday's Law. The lower the value of 'dt', the more will be the emf induced. But lower value of 'dt' means that core is moving faster. Thus we can say that the faster the movement of core, the greater will be the magnitude of emf induced in secondary windings. To get a single output voltage from the Linear Variable Differential Transformer (LVDT), both the secondary winding are connected in series but in phase opposition as shown in figure below.



Due to this connection, the net output voltage E_0 of the LVDT is given as below.

$$E_0 = E_{s1} - E_{s2}$$

Since the secondary windings of LVDT are identical and placed symmetrically on either side of core, therefore under normal position the flux linkage of both the secondary winding S_1 & S_2 will be same. This means $E_{s1} = E_{s2}$ and hence net output voltage E_0 of LVDT = 0. This position of soft iron core is called NULL position. Thus NULL position of Linear Variable Differential Transformer is the normal position of movable core where the net output voltage is zero.

19b

Write short notes on

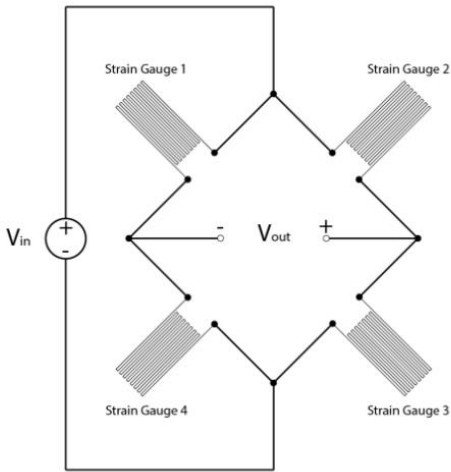
. I) Digital Multi Meter and II. Clamp on meter

Ans

A digital multimeter is a test tool used to measure two or more electrical values—principally voltage (volts), current (amps) and resistance (ohms). It is a standard diagnostic tool for technicians in the electrical/electronic industries. Digital multimeters long ago replaced needle-based analog meters due to their ability to measure with greater accuracy, reliability and increased impedance. Fluke introduced its first digital multimeter in 1977.

Digital multimeters combine the testing capabilities of single-task meters—the voltmeter (for measuring volts), ammeter (amps) and ohmmeter (ohms). Often, they include several

	additional specialized features or advanced options. Technicians with specific needs, therefore, can seek out a model targeted to meet their needs. The face of a multimeter typically includes four components: • Display: Where measurement readouts can be viewed. • Buttons: For selecting various functions; the options vary by model. • Dial (or rotary switch): For selecting primary measurement values (volts, amps, ohms). • Input jacks: Where test leads are inserted. Test leads are flexible, insulated wires (red for positive, black for negative) that plug into the DMM. They serve as the conductor from the item being tested to the multimeter. The probe tips on each lead are used for testing circuits. The terms counts and digits are used to describe a digital multimeter's resolution—how fine a measurement a meter can make. By knowing a multimeter's resolution, a technician can determine if it is possible to see a small change in a measured signal. Clamp on Meter A clamp meter is an electrical test tool that combines a basic digital multimeter with a current sensor. Clamps measure current. Probes measure voltage. Having a hinged jaw integrated into an electrical meter allows technicians to clamp the jaws around a wire, cable or other conductor at any point in an electrical system, then measure current in that circuit without disconnecting/deenergizing it. Beneath their plastic moldings, hard jaws consist of ferrite iron and are engineered to detect, concentrate and measure the magnetic field being generated by current as it flows through a conductor. 1. Current-sensing jaw. 2. Tactile barrier (to protect fingers from shocks). 3. Hold button: Freezes the display reading. Reading is released when button is pushed a second time. 4. Dial (aka rotary switch). 5. Display. 6. Backlight button. 7. Min Max button: On first push, display shows maximum input. On subsequent pushes, minimum and average inputs are shown. Works in current, voltage and frequency modes. 8. Inrush current button. 9. Zero button (yellow): Removes dc offset from dc current measurements. Also serves as dial's shift button to select yellow functions scattered around the dial. 10. Jaw release lever. 11. Alignment marks: To meet accuracy specifications, a conductor must be aligned with these marks. 12. Common input jack. 13. Volts/ohm input jack. 14. Input for flexible current probe. Originally created as a single-purpose test tool, modern clamp meters offer more measurement functions, greater accuracy and in some cases specialized measurement features. Today's clamp
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	meters include most of the basic functions of a digital multimeter (DMM), such as the ability to measure voltage, continuity and resistance. Clamp meters have become popular tools primarily for two reasons: • Safety. Clamp meters allow electricians to bypass the old-school method of cutting into a wire and inserting a meter's test leads into the circuit to take an in-line current measurement. The jaws of a clamp meter do not need to touch a conductor during a measurement. • Convenience. During a measurement, it is not necessary to shut off the circuit carrying current—a big boost in efficiency. Clamp meters are preferred for measuring high levels of current. DMMs cannot measure 10 A of current for more than 30 seconds without risking damage to the meter.
20a	How strain is measured using a strain gauge?
Ans	Now that the process of strain measurements has been established, the next step when using strain gauges is getting useful data. The strain gauge must be connected to an electrical circuit that is capable of accurately responding to the minute changes in resistance associated with strain. Multiple strain gauges can be used in a divided bridge circuit to measure small changes in electrical resistance. This is called a Wheatstone bridge. In a Wheatstone bridge configuration, an excitation voltage is applied across the circuit, and the output voltage is measured across two points in the middle of the bridge. When there is no load acting on the load cell, the Wheatstone bridge is balanced and there is zero output voltage. Any small change in the material under the strain gauge results in a change in the resistance of the strain gauge as it deforms with the material. This causes the bridge to be thrown out of balance, resulting in a change in the output voltage. As stated earlier, the resistance change is minute, which means that signal amplification is often needed to properly determine changes. The amplification process strengthens the strain signal changes; however, it also leads to more unwanted noise also being detected in the signal. Signal conditioning filters out the excess noise, ensuring accurate and understandable data.  The diagram illustrates a Wheatstone bridge circuit. It consists of four strain gauges, labeled Strain Gauge 1, Strain Gauge 2, Strain Gauge 3, and Strain Gauge 4, arranged in a diamond shape. An input voltage V_{in} is applied across the top and bottom nodes of the bridge. The output voltage V_{out} is measured across the left and right nodes of the bridge. The bridge is shown in a balanced state, with V_{out} indicated as zero.
20 b	Draw a neat block diagram of a Cathode Ray Oscilloscope and describe the function of each block in detail.

Ans

The cathode-ray oscilloscope (CRO) is a common laboratory instrument that provides accurate time and amplitude measurements of voltage signals over a wide range of frequencies. Its reliability, stability, and ease of operation makes it suitable as a general purpose laboratory instrument. Figure shows the basic block diagram of a general purpose CR oscilloscope.

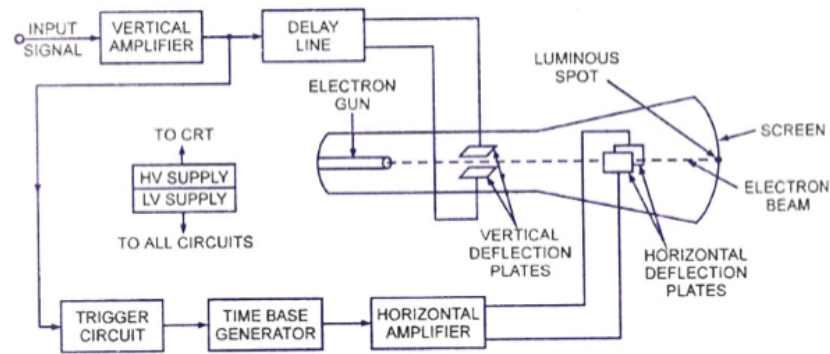


Figure - Block Diagram of General Purpose CRO

A general purpose oscilloscope consists of the following parts:

Cathode ray tube

Vertical amplifier

Delay line

Time base generator

Horizontal amplifier

Trigger circuit

Power supply

Cathode Ray Tube - It is the heart of the oscilloscope. When the electrons emitted by the electron gun strikes the phosphor screen, a visual signal is displayed on the CRT.

Vertical Amplifier - The input signals are amplified by the vertical amplifier. Usually, the vertical amplifier is a wide band amplifier which passes the entire band of frequencies.

Delay Line - As the name suggests, this circuit is used to delay the signal for a period of time in the vertical section of CRT. The input signal is not applied directly to the vertical plates because the part of the signal gets lost, when the delay time is not used. Therefore, the input signal is delayed by a period of time.

Time Base (Sweep) Generator - Time base circuit uses a uni-junction transistor, which is used to produce the sweep. The saw tooth voltage produced by the time base circuit is required to deflect the beam in the horizontal section. The spot is deflected by the saw tooth voltage at a constant time dependent rate.

Horizontal Amplifier - The saw tooth voltage produced by the time base circuit is amplified by the horizontal amplifier before it is applied to horizontal deflection plates.

Trigger Circuit - The signals which are used to activate the trigger circuit are converted to trigger pulses for the precision sweep operation whose amplitude is uniform. Hence input signal and the sweep frequency can be synchronized.

Power supply - The voltages required by CRT, horizontal amplifier, and vertical amplifier are provided by the power supply block. It is classified into two types -

(1) Negative high voltage supply

(2) Positive low voltage supply

The voltage of negative high voltage supply is from -1000V to -1500V. The range of positive voltage supply is from 300V to 400V.

	Functioning – The cathode ray is a beam of electrons which are emitted by the heated cathode (negative electrode) and accelerated toward the fluorescent screen. The assembly of the cathode, intensity grid, focus grid, and accelerating anode (positive electrode) is called an electron gun. Its purpose is to generate the electron beam and control its intensity and focus. Between the electron gun and the fluorescent screen are two pairs of metal plates - one oriented to provide horizontal deflection of the beam and one pair oriented to give vertical deflection to the beam. These plates are thus referred to as the horizontal and vertical deflection plates. The combination of these two deflections allows the beam to reach any portion of the fluorescent screen. Wherever the electron beam hits the screen, the phosphor is excited and light is emitted from that point. This conversion of electron energy into light allows us to write with points or lines of light on an otherwise darkened screen. The linear deflection or sweep of the beam horizontally is accomplished by use of a sweep generator that is incorporated in the oscilloscope circuitry. In the most common use of the oscilloscope, the signal to be studied is first amplified and then applied to the vertical (deflection) plates to deflect the beam vertically, and at the same time, a voltage that increases linearly with time is applied to the horizontal (deflection) plates, thus causing the beam to be deflected horizontally at a uniform rate. The signal applied to the vertical plates is thus displayed on the screen as a function of time. The horizontal axis serves as a uniform time scale. Role of sweep in CRO – The sweep, also known as saw tooth pulse, is required to deflect the beam in the horizontal section. To obtain steady traces on the tube face, an internal number of cycles of the unknown signal must be associated with each cycle of the sweep generator. Thus, with such a matching of synchronization of the two deflections, the pattern on the tube face repeats itself and hence appears to remain stationary. The persistence of vision of the human eye and of the glow of the fluorescent screen aids in producing a stationary pattern. In simple words, the sweep is the horizontal speed of the cathode ray tube's spot which is used to create a trace. This ensures that the signal being tested is locked on the screen and does not drift.
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